

Biliary, Renal, Neurologic, and Skeletal Syndrome (BRENS): A Comprehensive Disease Characteristics Report

Disease: Biliary, Renal, Neurologic, and Skeletal Syndrome (BRENS) **MONDO ID:** MONDO:0859191 · **OMIM:** #619534 · **Category:** Mendelian (autosomal recessive) **Causal gene:** IFT56 (formerly TTC26; synonym DYF13), HGNC:21882, chr7q34

Summary

Biliary, Renal, Neurologic, and Skeletal Syndrome (BRENS) is an ultra-rare, autosomal-recessive, multisystem **ciliopathy** caused by biallelic loss-of-function variants in **IFT56 (formerly TTC26)**, a tetratricopeptide-repeat (TPR) protein that is a core component of the **intraflagellar transport (IFT)-B complex**. IFT56/TTC26 is required for the anterograde transport of a selective set of cargoes along the ciliary axoneme; its loss compromises the function of both primary (signaling) cilia and motile cilia. The disorder was first delineated in 2020 by Shaheen et al., who described seven individuals from seven families with a syndromic ciliopathy featuring severe neonatal cholestasis, and identified three different homozygous *TTC26* variants that fully segregated with the phenotype [PMID: 31595528].

Mechanistically, IFT56 loss dysregulates **ciliary Sonic Hedgehog (Shh)–GLI signaling** during embryonic development. Work in the *Ttc26* hop-sterile mouse pinpointed the defect to failure of **GLI dissociation from its negative regulator SUFU** at the ciliary tip, downstream of normal GLI accumulation [PMID: 25340710]. A more recent study adds that TTC26 is required for ciliary localization of the methyltransferase **PRMT7**, which methylates **GLI2** to maintain Shh–GLI2 signaling [PMID: 42178579]. Because Hedgehog signaling patterns the limb, neural tube, skeleton, and multiple viscera, and because cilia are essential in cholangiocytes, renal tubular cells, ependyma, and photoreceptors, the consequence is a congenital, variably expressed multi-organ malformation syndrome.

Clinically, BRENS produces **neonatal cholestasis / fibrocystic biliary disease** (potentially lethal or requiring liver transplant), **renal cysts/dysplasia**, **brain malformations** (hydrocephalus, aqueductal stenosis), **pituitary stalk interruption syndrome (PSIS) with hypopituitarism and diabetes insipidus**, **polydactyly/syndactyly** and other skeletal anomalies, plus cardiac, ocular (optic atrophy), and craniofacial (cleft lip/palate) features. Expressivity is variable — one Chinese patient had renal, neurologic, and skeletal features **without** biliary involvement [PMID: 39514123]. Diagnosis is molecular (WES/WGS identifying biallelic *IFT56/TTC26* variants); there is **no curative therapy**, and management is supportive and multidisciplinary. Prognosis is guarded, dominated by the severity of the neonatal hepatic phenotype. Fewer than ~25 patients have been published worldwide as of 2025.

Key Findings

1. BRENS is an autosomal-recessive ciliopathy caused by biallelic *IFT56/TTC26* variants

The disease-gene relationship is firmly established. In the foundational cohort, whole-exome sequencing of seven individuals from seven families with syndromic ciliopathy features — including severe neonatal cholestasis — revealed *"three different homozygous variants in Tetratricopeptide Repeat Domain 26 (TTC26) that fully segregated with the phenotype"* after positional mapping to a single locus on chromosome 7q [PMID: 31595528]. Multiple independent families have since confirmed the gene: a homozygous splice-site variant c.4-1G>C [PMID: 34177428]; the recurrent homozygous missense c.695A>G (p.Asn232Ser) [PMID: 32617964]; a homozygous intronic c.1006-5T>C [PMID: 38135897]; and compound-heterozygous alleles in a Chinese boy [PMID: 39514123]. Inheritance is autosomal recessive, and several families are consanguineous. The disease carries OMIM #619534; the gene carries OMIM 617453.

"Whole-exome sequencing revealed three different homozygous variants in Tetratricopeptide Repeat Domain 26 (TTC26) that fully segregated with the phenotype." — [PMID: 31595528]

"We describe seven individuals from seven families with syndromic ciliopathy clinical features, including severe neonatal cholestasis (lethal in one and necessitating liver transplant in two)." — [PMID: 31595528]

2. TTC26/IFT56 is a conserved IFT-B core component required for cargo-selective transport

TTC26 (also called DYF13 in *Chlamydomonas* and *C. elegans*) is an integral component of the intraflagellar transport complex B. Ishikawa et al. showed that "*TTC26/DYF13 is an IFT complex B protein in mammalian cells and Chlamydomonas reinhardtii*" [PMID: 24596149]. Loss of TTC26 produces short cilia with abnormal motility while leaving IFT-particle assembly and speed largely normal; critically, "*a particular set of proteins involved in motility was specifically depleted in the dyf13 mutant*" — evidence that TTC26 mediates **cargo-selective transport** rather than bulk IFT [PMID: 24596149]. Structural/biochemical studies place TTC26/IFT56 in the **IFT-B core subcomplex** [PMID: 26980730] and show it dimerizes with and directly binds **IFT46** [PMID: 27927754, 25340710]. Xin et al. demonstrated that "*IFT56 regulates vertebrate developmental patterning by maintaining IFTB complex integrity and ciliary microtubule architecture*" [PMID: 28264835].

3. Mechanism: TTC26 loss dysregulates Sonic Hedgehog signaling via impaired GLI-SUFU dissociation and GLI2 methylation

The core pathogenic mechanism is disruption of ciliary Hedgehog signal transduction. In the spontaneous *Ttc26* hop-sterile mouse, the group found that "*the hop mutation is located in the Ttc26 gene and impairs Hedgehog (Hh) signaling*" [PMID: 25340710]. Importantly, the defect is not a failure of cilium formation — cilia number and length are preserved — but a downstream signaling failure: "*hop did not interfere with Hh-induced accumulation of Gli at the tip of the primary cilium, but rather with the subsequent dissociation of Gli from its negative regulator, SuFu*" [PMID: 25340710]. Patient-derived cells show cilia of variable length with dysregulated Sonic Hedgehog signaling and abnormal IFT-B staining [PMID: 31595528]. A recent study adds a molecular refinement: "*TTC26 is required for the localization of protein arginine methyltransferase 7 (PRMT7) to the primary cilium, enabling methylation of GLI2*" — with FLNB, to maintain Shh–GLI2 signaling [PMID: 42178579].

4. Phenotypic spectrum spans biliary, renal, neurologic, skeletal, endocrine, cardiac, ocular, and craniofacial systems

The syndrome is defined by multi-organ involvement with variable expressivity. Core features from the founding cohort include severe neonatal cholestasis (lethal in one, requiring liver transplant in two) and fibrocystic liver/biliary disease [PMID: 31595528]. Pituitary involvement is a recurrent — and now recognized as characteristic — feature: four patients with

homozygous p.Asn232Ser had **pituitary stalk interruption syndrome (PSIS)**, delineated "*as a novel clinical feature of this disorder*" [PMID: 32617964]. Across cases, reported manifestations include "*cholestasis, cystic dilatation of intrahepatic biliary ducts, diabetes insipidus, dysmorphic facial features, optic atrophy, pituitary hypoplasia, hydrocephalus, aqueductal stenosis, hyperextensible knee joints, bilateral knee dislocation, polydactyly, and syndactyly*" [PMID: 34177428], plus cleft lip/palate with probable hearing loss [PMID: 38135897]. Demonstrating the breadth of expressivity, a Chinese boy showed renal, neurologic, and skeletal features but no biliary disease — "*the first description of BRENS syndrome without biliary involvement*" [PMID: 39514123].

System	Key phenotypes	HPO suggestion	Representative PMID
Hepatobiliary	Neonatal cholestasis, fibrocystic biliary disease, ductal plate malformation, intrahepatic duct dilatation	HP:0011967, HP:0001395	31595528, 34177428
Renal/urinary	Renal cysts, dysplasia, nephropathy	HP:0000107	31595528, 38135897
Nervous/CNS	Hydrocephalus, aqueductal stenosis	HP:0000238, HP:0002410	31595528, 34177428
Endocrine (pituitary)	PSIS, hypopituitarism, diabetes insipidus	HP:0010626, HP:0000873	32617964, 38135897
Skeletal/limb	Pre/postaxial polydactyly, syndactyly, joint hyperextensibility/dislocation	HP:0100259, HP:0001388	34177428
Ocular	Optic atrophy	HP:0000648	34177428
Cardiac	Congenital heart defect	HP:0001627	38135897
Craniofacial	Cleft lip/palate, dysmorphism, possible hearing loss	HP:0000175	38135897

5. Cross-species model organisms robustly recapitulate the ciliopathy

Multiple model systems reproduce BRENS-relevant phenotypes and have illuminated mechanism. **Mouse:** the spontaneous *Ttc26* hop-sterile mutant is "*characterized by a hopping gait, polydactyly, hydrocephalus, and male sterility*" with inner dynein-arm deficiency, absent sperm flagella, and impaired Hedgehog signaling [PMID: 25340710]; a 2025 study found that

Ift56 loss-of-function "has dramatic phenotypic differences depending on the genetic background in mice", establishing genetic-background modifiers [PMID: 41352382]. **Zebrafish:** morpholino knockdown of *ttc26* "caused ciliary defects in the pronephric kidney at 27 h postfertilization and distension/dilation of pronephros" [PMID: 22718903], plus photoreceptor outer-segment defects [PMID: 36533556]. **Chlamydomonas:** *dyf13/TTC26* mutation gives short flagella with abnormal motility and selective loss of motility proteins [PMID: 24596149]. A double *Flnb;Ttc26* heterozygous mouse models adolescent idiopathic scoliosis via Shh–GLI2 [PMID: 42178579].

6. IFT56/TTC26 is LoF-tolerant in heterozygotes; BRENS is ultra-rare

gnomAD v4 constraint metrics for IFT56/TTC26 (ENSG00000105948) indicate that heterozygous loss-of-function is **tolerated** — $pLI \approx 3 \times 10^{-10}$, observed/expected LoF (o/e) = 0.61 (90% CI 0.49–0.77; LOEUF 0.77), with 52 observed vs 85 expected pLoF variants. This tolerance is exactly what is expected for an autosomal-recessive disease gene in which carriers are unaffected. Summing 133 high-confidence pLoF allele frequencies yields a cumulative pLoF allele frequency of $\approx 2.36 \times 10^{-4}$, implying a carrier frequency of roughly **1 in ~2,100** for truncating alleles alone, and a predicted homozygous/compound-het birth frequency from pLoF alleles of $\approx 5.6 \times 10^{-8}$ (~1 in 18 million). Because most reported disease alleles are splice-region/missense (e.g., the recurrent c.695A>G, p.Asn232Ser) rather than canonical LoF, the true prevalence is higher than the pLoF-only estimate but remains ultra-rare (<25 published patients worldwide as of 2025).

7. Diagnosis is genetic; management is supportive with guarded prognosis

BRENS diagnosis is established by molecular genetic testing — **whole-exome or whole-genome sequencing** identifying biallelic *IFT56/TTC26* variants (the original cohort combined positional mapping with WES [PMID: 31595528]; a later family used WGS with Sanger confirmation [PMID: 34177428]). Supportive workup findings include neonatal conjugated hyperbilirubinemia with elevated liver enzymes; liver biopsy showing ductal plate malformation/biliary fibrosis; brain MRI showing pituitary stalk interruption (thin/absent stalk, ectopic posterior pituitary, anterior pituitary hypoplasia) and hydrocephalus/aqueductal stenosis; renal imaging showing cysts/dysplasia; and an endocrine panel revealing hypopituitarism and diabetes insipidus. **No disease-specific or curative therapy exists.** Management is supportive and organ-directed: ursodeoxycholic acid and nutritional support for cholestasis; **liver transplantation** for end-stage liver disease (2 of 7 original patients required transplant, 1 died — "severe neonatal cholestasis (lethal in one and necessitating liver transplant in two)" [PMID: 31595528]); multi-hormone replacement for pituitary insufficiency, where "hormonal replacement therapy with hydrocortisone, levothyroxine, and growth hormone

led to clinical stabilization" [PMID: 42460215]; ventriculoperitoneal shunting for hydrocephalus; and surgical correction of polydactyly and clefts. Prognosis is guarded; severe neonatal cholestasis can be lethal.

8. Identifiers and IFT56 protein architecture

Disease identifiers: **MONDO:0859191** ("biliary, renal, neurologic, and skeletal syndrome"); **OMIM #619534**; **MedGen C1794200**; **UMLS C5561990**; MalaCards entry present. No dedicated Orphanet ORPHA code or specific ICD-10/ICD-11/MeSH term is currently mapped (classified broadly under ciliopathy/congenital malformation syndromes). Gene: **IFT56** (formerly TTC26; synonym DYF13), **HGNC:21882**, **OMIM 617453**, Ensembl **ENSG00000105948**, chr7q34. Protein: **UniProt A0AVF1**, "Intraflagellar transport protein 56," 554 aa, containing **four tetratricopeptide-repeat (TPR) motifs** (aa 57–90, 92–125, 151–184, 468–501). The recurrent founder missense p.Asn232Ser lies in the inter-repeat region; splice-site variants predominate among disease alleles.

9. Anatomical, cellular, and subcellular map

Documented across cases [PMIDs 31595528, 34177428, 32617964, 38135897, 39514123], with ontology suggestions:

Level	Structure / cell	Ontology term
Hepatobiliary	Bile duct / liver; cholangiocyte	UBERON:0002394 / UBERON:0002107; CL:0002326
Renal	Kidney tubule/collecting duct; renal epithelial cell	UBERON:0002113; CL:1000454
Nervous	Cerebral aqueduct; ependymal cell	UBERON:0002289; CL:0000065
Endocrine	Pituitary gland; neurohypophysis	UBERON:0000007; UBERON:0002590
Eye	Photoreceptor layer; photoreceptor cell	UBERON:0001789; CL:0000210
Skeletal	Digit (autopod)	UBERON:0002389
Cardiac	Heart	UBERON:0000948
Craniofacial	Mouth (lip/palate)	UBERON:0000165
Subcellular	Cilium; 9+2 motile cilium; ciliary tip; IFT particle B	GO:0005929; GO:0097729; GO:0097542; GO:0030992

Subcellular localization is confirmed: *"We localized Ttc26 to the transition zone of photoreceptor and to the transition zone of cilia in cultured murine inner medullary collecting duct 3 (mIMCD3) renal cells"* [PMID: 22718903]. Tissue expression supports hepatobiliary primacy: *"strong expression of Ttc26 in the embryonic mouse liver in a pattern consistent with its proposed role in the normal development of the intrahepatic biliary system"* [PMID: 31595528].

10. Natural history and variant landscape

BRENS is **congenital/neonatal in onset** (cholestasis and malformations present at or near birth), **chronic and lifelong**, without spontaneous remission. The hepatic component is often **progressive** (cholestasis → biliary fibrosis → end-stage liver disease/transplant), whereas malformations (polydactyly, hydrocephalus, PSIS) are static-congenital and endocrine deficiency is stable but permanent [PMID: 31595528, 32617964]. Critical intervention windows: the embryonic period fixes malformations (irreversible), and the neonatal period is critical for life-saving hormone replacement and cholestasis/transplant management. The variant landscape comprises splice-region and missense loss-of-function changes (c.4-1G>C, c.695A>G p.Asn232Ser, c.1006-5T>C, c.1069+5G>A, c.511A>G, c.1099T>C, plus three homozygous founding-

cohort variants); ClinVar lists 200+ submissions, predominantly population VUS/benign, consistent with a recessive, LoF-tolerant gene. Etiology is **exclusively genetic** (autosomal recessive); no environmental, infectious, epigenetic, or chromosomal mechanism is implicated.

Section-by-Section Report

1. Disease Information

BRENS is an ultra-rare autosomal-recessive multisystem ciliopathy affecting biliary, renal, neurologic, and skeletal systems (plus endocrine, cardiac, ocular, and craniofacial). Key identifiers: MONDO:0859191, OMIM #619534, MedGen C1794200, UMLS C5561990. No dedicated Orphanet/ICD/MeSH code is currently mapped. Synonyms: "TTC26 ciliopathy," "biliary ciliopathy (TTC26-related)." Information is derived from **aggregated disease-level resources and published individual case reports** (not EHR cohorts) — fewer than ~25 patients worldwide.

2. Etiology

The primary and sole established cause is **genetic**: biallelic loss-of-function variants in IFT56/TTC26. No environmental, infectious, lifestyle, or toxic contributing factors are implicated. Genetic risk requires two pathogenic alleles (autosomal recessive); consanguinity is a strong risk factor (several reported families are consanguineous, and homozygous founder alleles such as p.Asn232Ser recur). No protective variants or gene–environment interactions are described. Modifier effects are documented at the model-organism level: genetic background dramatically alters phenotype severity in *Ift56/Ttc26* mutant mice [PMID: 41352382], and FLNB acts as a genetic modifier of the Shh–GLI2 axis [PMID: 42178579].

3. Phenotypes

See Finding 4 table. Onset is neonatal/congenital. Severity is variable (from lethal neonatal cholestasis to milder biliary-sparing presentations). Progression: hepatic disease progressive; malformations static; endocrine deficits permanent. Quality-of-life impact is severe where hypopituitarism, hydrocephalus, and liver disease coexist, requiring lifelong hormone replacement and organ-directed care. Suggested HPO terms: Neonatal cholestasis (HP:0011967), Hepatic fibrosis (HP:0001395), Polydactyly (HP:0100259), Hydrocephalus

(HP:0000238), Aqueductal stenosis (HP:0002410), Anterior pituitary hypoplasia (HP:0010626), Diabetes insipidus (HP:0000873), Optic atrophy (HP:0000648), Renal cyst (HP:0000107), Cleft lip/palate (HP:0000175), Congenital heart defect (HP:0001627), Joint hyperlaxity (HP:0001388).

4. Genetic/Molecular Information

Causal gene: IFT56/TTC26 (HGNC:21882, OMIM 617453). Variant types: predominantly splice-site and missense (loss-of-function). Classification per ACMG/AMP: reported disease alleles are pathogenic/likely pathogenic; the gene shows abundant benign/VUS population variation. Allele frequencies in gnomAD are consistent with an ultra-rare recessive disorder (see Finding 6). Origin is germline. Functional consequence is loss of function (disrupted IFT-B integrity and cargo transport). Modifier genes: FLNB and genetic background (from models). No epigenetic or chromosomal mechanism is implicated.

5. Environmental Information

Not applicable — BRENS is a purely Mendelian disorder. No environmental, lifestyle, or infectious agents contribute.

6. Mechanism / Pathophysiology

Ordered causal chain: 1. Biallelic loss-of-function variants in **IFT56/TTC26** → loss of functional IFT56 protein. 2. Loss of IFT56 → **destabilization of the IFT-B core subcomplex** and impaired ciliary microtubule architecture [PMID: 28264835] → **cargo-selective failure of anterograde intraflagellar transport** (motility-related and signaling cargoes) [PMID: 24596149]. 3. Impaired IFT → **dysfunctional primary (signaling) and motile cilia** across tissues (short/variable-length cilia; abnormal motility) [PMID: 31595528, 24596149]. 4a. In primary cilia → failure of **GLI dissociation from SUFU** at the ciliary tip (inferred to also involve loss of ciliary PRMT7 → reduced GLI2 methylation) → **dysregulated Sonic Hedgehog–GLI signaling** [PMID: 25340710, 42178579]. 4b. In motile cilia → inner dynein-arm deficiency and impaired ciliary/flagellar motility → ependymal/CSF-flow and reproductive defects (demonstrated in mouse) [PMID: 25340710]. 5. Dysregulated Shh–GLI patterning during embryogenesis → **abnormal development of limb (polydactyly), neural tube/brain (hydrocephalus, aqueductal stenosis), pituitary (PSIS), intrahepatic biliary tree (ductal plate malformation/cholestasis), kidney (cysts), heart, eye, and craniofacial structures**. 6. Postnatally → progressive biliary fibrosis and end-stage liver disease; permanent hypopituitarism and diabetes insipidus; static malformations → **the BRENS clinical phenotype**.

Molecular pathways: **Hedgehog/GLI** (central). Cellular processes: ciliogenesis, intraflagellar transport, developmental patterning. Protein dysfunction: loss of function of an IFT-B TPR scaffold destabilizing the complex. Suggested GO terms: intraciliary transport (GO:0042073), smoothened signaling pathway (GO:0007224), cilium assembly (GO:0060271), determination of left/right symmetry (GO:0007368). Suggested CL terms: cholangiocyte (CL:0002326), ependymal cell (CL:0000065), photoreceptor cell (CL:0000210), kidney epithelial cell (CL:1000454).

7. Anatomical Structures Affected

See Finding 9. Primary organs: liver/intrahepatic bile ducts, kidney, brain (ventricular system), pituitary. Secondary/associated: heart, eyes, skeleton/limbs, craniofacial structures. Involvement is generally bilateral. Subcellular: primary and motile cilia (axoneme, transition zone, ciliary tip), IFT-B particle.

8. Temporal Development

Onset congenital/neonatal; course chronic-lifelong. Hepatic disease progressive; malformations static; endocrine deficits permanent. No remission. Critical windows: embryonic (malformation fixation, irreversible) and neonatal (life-saving hormone replacement, cholestasis/transplant management).

9. Inheritance and Population

Autosomal recessive. Ultra-rare (<25 published patients; estimated carrier frequency ~1/2,000 for truncating alleles). Penetrance appears complete for biallelic pathogenic genotypes, with **variable expressivity** (biliary-sparing cases exist [PMID: 39514123]). Consanguinity and founder alleles (p.Asn232Ser) contribute. No confirmed sex bias in the syndrome itself (male sterility is seen in the mouse model). No genetic anticipation (not a repeat-expansion disorder).

10. Diagnostics

Molecular diagnosis by WES/WGS (biallelic IFT56/TTC26 variants). Supportive: conjugated hyperbilirubinemia, elevated liver enzymes; liver biopsy (ductal plate malformation/fibrosis); brain MRI (PSIS triad, hydrocephalus/aqueductal stenosis); renal ultrasound (cysts/dysplasia); endocrine panel (hypopituitarism, diabetes insipidus). Differential diagnosis: other syndromic ciliopathies (Meckel, Joubert, Bardet-Biedl, nephronophthisis-related; TTC12/TTC21B multisystem ciliopathies [PMID: 36273201]), Alagille syndrome, and other causes of neonatal cholestasis with malformations. Screening: cascade carrier testing in families; prenatal/preimplantation testing where the familial variant is known.

11. Outcome/Prognosis

Guarded. Neonatal cholestasis can be lethal; liver transplantation may be required (2/7 transplanted, 1 death in the founding cohort [PMID: 31595528]). Survivors face lifelong morbidity from hypopituitarism, diabetes insipidus, hydrocephalus, and organ malformations. No formal survival statistics exist given the small patient numbers. Prognostic factors: severity of the hepatic phenotype and presence/absence of biliary involvement.

12. Treatment

No curative/disease-specific therapy. Supportive/organ-directed: ursodeoxycholic acid and nutritional support (cholestasis); liver transplantation (end-stage liver disease); multi-hormone replacement — hydrocortisone, levothyroxine, growth hormone, desmopressin for DI [PMID: 42460215, 40539145]; VP shunt (hydrocephalus); surgical correction of polydactyly and cleft lip/palate. Suggested NCIT terms: Liver Transplantation (NCIT:C15238), Hormone Replacement Therapy (NCIT:C15667), Ursodeoxycholic Acid (NCIT:C29273), Ventriculoperitoneal Shunt (NCIT:C50124). No pharmacogenomic, gene, cell, or RNA therapies are established or in trials for BRENS.

13. Prevention

No primary prevention beyond **genetic counseling** for at-risk (especially consanguineous) families. Secondary/tertiary prevention: early diagnosis enabling timely hormone replacement, cholestasis management, and shunting to prevent complications. Options where the familial variant is known: carrier screening, prenatal testing, preimplantation genetic diagnosis. No immunization or environmental intervention applies.

14. Other Species / Natural Disease

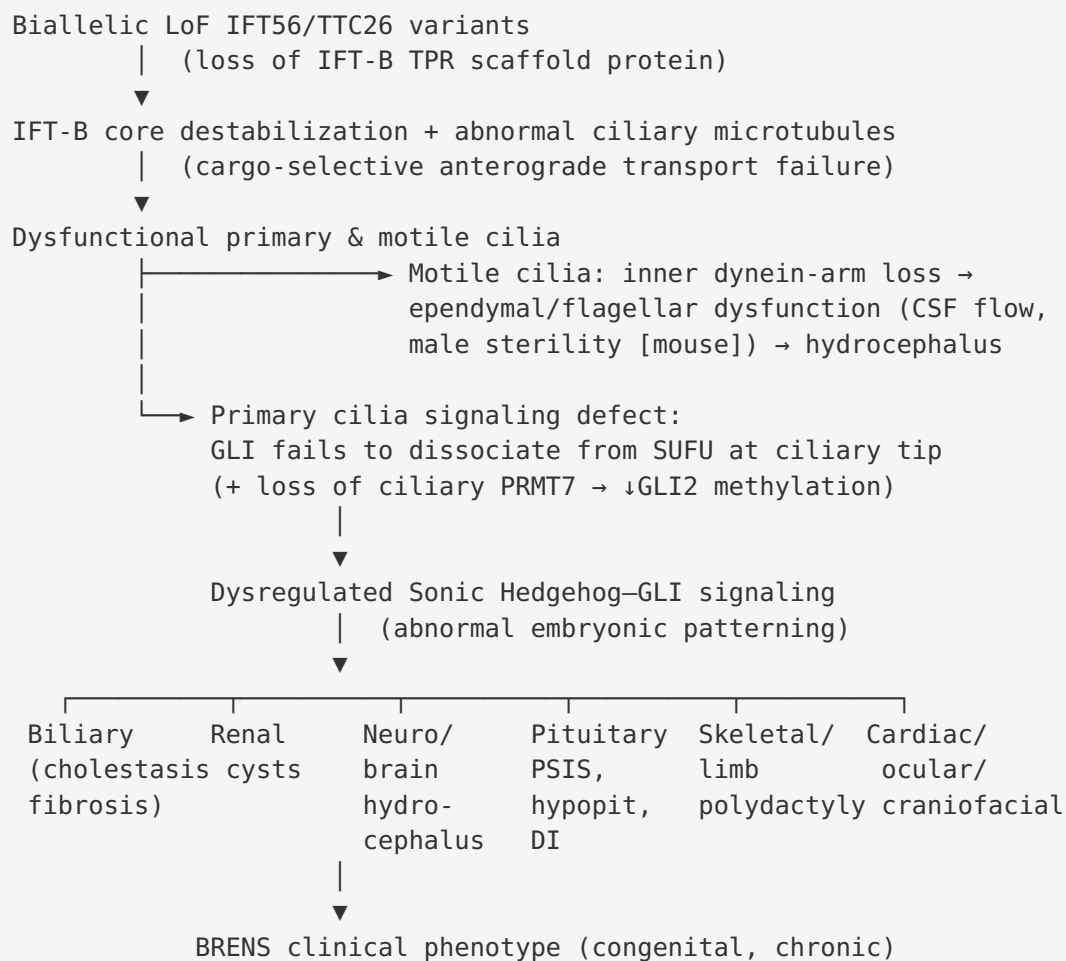
No naturally occurring companion-animal or wildlife disease is reported (no OMIA entry noted). Orthologous genes: mouse *Ttc26*, zebrafish *ttc26*, *Chlamydomonas* DYF13, *C. elegans* dyf-13. Evolutionary conservation of IFT-B and its function is high across ciliated eukaryotes. No zoonotic potential (Mendelian disorder).

15. Model Organisms

Robust models exist (Finding 5): mouse (*Ttc26* hop-sterile spontaneous mutant; *Ift56/Ttc26* engineered LoF with background-dependent severity; *Flnb;Ttc26* double heterozygote for scoliosis), zebrafish (*ttc26* morpholino/CRISPR), and *Chlamydomonas* (*dyf13*). Phenotype

recapitulation is strong for polydactyly, hydrocephalus, renal cystic/pronephric defects, photoreceptor defects, motile-cilia/flagellar dysfunction, and Hedgehog-signaling readouts. Limitations: the severe human biliary phenotype is incompletely modeled; genetic background strongly modulates murine phenotypes, complicating cross-study comparison. Resources: MGI, ZFIN, IMPC.

Mechanistic Model (Diagram)



Evidence Base

PMID	Title (abbrev.)	Contribution
31595528	<i>Biallelic Mutations in TTC26 (IFT56) Cause Severe Biliary Ciliopathy</i>	Foundational: defines disease, gene, recessive segregation, biliary severity
25340710	<i>A mutation in mouse ttc26 leads to impaired hedgehog signaling</i>	Core mechanism: GLI–SUFU dissociation failure; mouse model
42178579	<i>FLNB and TTC26 regulate ciliary Hedgehog signaling...</i>	Mechanistic refinement: PRMT7→GLI2 methylation; FLNB modifier
24596149	<i>TTC26/DYF13 is an IFT protein required for transport of motility proteins</i>	Establishes IFT-B membership; cargo selectivity
28264835	<i>IFT56 regulates vertebrate patterning...</i>	IFT-B integrity, microtubule architecture
26980730	<i>Overall architecture of the IFT-B complex</i>	Places TTC26/IFT56 in IFT-B core
27927754	<i>ARL13B/INPP5E regulate retrograde trafficking</i>	IFT46–IFT56 dimer; Hedgehog links
32617964	<i>PSIS broadens the TTC26 ciliopathy spectrum</i>	Adds PSIS; recurrent p.Asn232Ser
34177428	<i>Identification of the c.4-1G>C variant...</i>	New family; enumerates multisystem phenotype
38135897	<i>Novel TTC26 variant...expansion of phenotype</i>	c.1006-5T>C; adds clefts, hearing loss
39514123	<i>BRENS in a Chinese boy</i>	Variable expressivity: biliary-sparing case
22718903	<i>Knockdown of ttc26 disrupts ciliogenesis...zebrafish</i>	Renal/photoreceptor model; transition-zone localization
36533556	<i>Variable phenotypes in zebrafish TZ mutants</i>	ttc26 crispant ciliary phenotype
41352382	<i>Genetic background influences Ift56/Ttc26 anomalies</i>	Genetic-background modifiers
42460215	<i>PSIS in a newborn with recurrent hypoglycemia</i>	Supports hormone replacement management
40539145	<i>PSIS: A Case Series</i>	PSIS clinical triad and management

PMID	Title (abbrev.)	Contribution
36273201	<i>TTC12/TTC21B multisystem ciliopathies</i>	Differential diagnosis / phenotypic overlap

Limitations and Knowledge Gaps

- **Very small patient population** (<25 published cases) limits confident estimates of penetrance, expressivity, sex ratio, survival, and genotype–phenotype correlation.
- **No formal epidemiology**: prevalence/incidence are estimated indirectly from gnomAD allele frequencies, not measured. The pLoF-only carrier estimate (~1/2,100) omits the missense/splice alleles that dominate the reported disease spectrum, so it underestimates true carrier frequency.
- **Incomplete mechanistic mapping** of how a single ciliary transport defect yields the specific organ set; the relative contributions of primary-cilia (Hedgehog) vs motile-cilia dysfunction to each organ phenotype are not fully resolved.
- **The severe human biliary phenotype is under-modeled** in animals; no model fully recapitulates neonatal cholestasis/biliary fibrosis.
- **No therapeutics** are disease-specific; no clinical trials exist. Pharmacogenomic and gene/cell/RNA therapy data are absent.
- **Ontology mapping gaps**: no Orphanet/ICD/MeSH code; ontology term suggestions here are proposed, not curated into the disease record.

Proposed Follow-up Experiments / Actions

1. **Establish a BRENS patient registry / GeneMatcher-linked cohort** to aggregate genotype–phenotype data, refine penetrance and expressivity, and capture natural history prospectively.
2. **Variant-level functional assays** (splice reporters for the recurrent intronic variants; rescue assays for missense alleles such as p.Asn232Ser) to firm up ACMG classification and reveal hypomorphic vs null effects underlying biliary-sparing presentations.

3. **Conditional/tissue-specific Ift56 mouse models** (hepatoblast/cholangiocyte, pituitary, renal) on a controlled genetic background to model the biliary phenotype and dissect organ-specific mechanism.
4. **Single-cell transcriptomics of patient-derived cholangiocyte and renal organoids/iPSC models** to map Hedgehog-target dysregulation and identify candidate therapeutic nodes (e.g., SMO agonists, PRMT7 modulation).
5. **Curate BRENS into Orphanet/ICD-11/MeSH and submit HPO/GO/CL/UBERON annotations** derived from this report to close ontology gaps.
6. **Systematic imaging/endocrine screening protocol** (neonatal MRI + endocrine panel) for any infant with cholestasis plus polydactyly to enable early hormone replacement during the critical neonatal window.

Evidence source types: primarily human clinical case reports/series and model-organism studies (mouse, zebrafish, Chlamydomonas), supplemented by in vitro biochemistry and computational (gnomAD constraint) analyses. All mechanistic and clinical claims are cited to primary literature by PMID.
